



PedestriGuard AI – Intelligent Pedestrian Detection and Behavior Analysis System

Project Report

*submitted in partial fulfillment of the
requirements for the award of the degree of*

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

by

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CANDIDATE'S DECLARATION

I/We hereby certify that the project work entitled **“PedestriGuard AI”** in partial fulfilment of the requirements for the award of the Degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING with specialization in AIML and submitted to the Department of Computer Science, School of Computer Science, University of Petroleum & Energy Studies, Delhi, is an authentic record of my/ our work carried out during a period from **JANUARY, 2026** to **MAY, 2026** under the supervision of Mrs. Vinamrata Rani.

The matter presented in this project has not been submitted by me/ us for the award of any other degree of this or any other University.

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Date: 29/04/2026

Mrs. Vinamrata Rani
Project Guide

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ABSTRACT

Due to rapid urbanization and growing smart cities, there is a need for pedestrian safety and intelligent surveillance. In public areas like streets, railway stations, airports, shopping complexes, and educational institutions, there is an endless stream of pedestrians, which makes it inefficient and cumbersome to manually watch them. Conventional surveillance systems work using CCTVs which only record video footage without interpreting any information from them or giving immediate warnings. In order to address this issue, a novel project named “PedestriGuard AI – Intelligent Pedestrian Detection and Behavior Analysis System” is conceived to solve this problem using Artificial Intelligence, Computer Vision, and Deep Learning.

The system proposed herein has a modular design and is capable of conducting multiple intelligent operations through one system. First, it utilizes object detection mechanisms to detect pedestrians in video streams. Second, it employs tracking techniques to track individuals by assigning unique identifiers. Third, the system leverages the technique of pose estimation to detect body posture and recognize the behavior of the individual whether walking, standing, sitting, running, or falling. Moreover, the system is capable of estimating demographic variables including age and gender. Such demographic features may be used for various purposes such as crowd behavior analysis, intelligent retailing systems, and urban planning.

From experiments conducted, we find that this technology is able to detect people accurately, predict age and gender without any errors, and perform behavior recognition quickly. Unlike the typical CCTV systems used for security purposes, PedestriGuard AI does more than just detecting people in an area; it identifies what activities the individuals have been engaged in. Therefore, such innovations will be of great importance in applications such as surveillance, healthcare, caring for the elderly, retail analytics, and public safety. In summary, through this experiment, it is evident that technology is transforming normal monitoring equipment into automated systems.

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1. INTRODUCTION

The rapid growth of urban populations and the expansion of transportation systems have significantly increased the complexity of pedestrian movement in modern cities. With more people using public spaces such as roads, transportation hubs, shopping centers, and public events, ensuring pedestrian safety and efficient crowd management has become a critical challenge. Traditional surveillance systems are mainly designed to record video footage or perform basic object detection, but they lack the ability to understand the behavior and context of pedestrian movement.

Recent advancements in artificial intelligence, particularly in computer vision and deep learning, have opened new opportunities for intelligent monitoring systems. These technologies enable machines to analyze visual data, identify objects, track movement patterns, and recognize activities in real time. Such capabilities are essential for developing advanced pedestrian monitoring solutions that can enhance situational awareness and support decision-making in smart city environments.

The PedestriGuard AI – Intelligent Pedestrian Detection and Behavior Analysis System is designed to address these challenges by combining multiple advanced technologies into a single integrated framework. The system uses deep learning algorithms to detect pedestrians, track their movement across video frames, analyze their behavior using pose estimation techniques, and extract demographic insights such as age group and gender. By transforming raw video data into meaningful information, the system can assist authorities, security agencies, and organizations in improving safety, efficiency, and operational intelligence.

This project aims to build a scalable and intelligent pedestrian monitoring system that can be deployed in a variety of real-world environments such as traffic intersections, public surveillance systems, healthcare facilities, and retail spaces. The integration of detection, tracking, and behavioral analysis makes the system capable of providing deeper insights than traditional monitoring solutions.

1.1 Background of the Problem

Due to urbanization and population explosion, public areas like roads, railways, airports, shopping complexes, educational institutions, and commercial zones have increased pedestrian traffic on a daily basis. Safety of pedestrians, crowd behavior analysis, and prevention of accidents have become difficult tasks for government bodies and institutions. Conventional surveillance systems depend

heavily on closed-circuit television cameras that capture the footage for post-processing purposes. Such systems need human supervision, which is cumbersome, less efficient, and highly susceptible to errors. They fail to analyze pedestrian movement and abnormal behavior in real-time and issue alerts when necessary. Hence, a demand for an automated system has emerged.

1.2 Need for Predictive Traffic Intelligence

The current monitoring system has become ineffective because of the fast decision-making required in the modern smart environment. An intelligent pedestrian monitoring system is capable of detecting people, tracking their movements, identifying abnormal activities, and recognizing any suspicious actions. Intelligent pedestrian monitoring systems will help in reducing the amount of work done by humans, increasing the efficiency of monitoring and also providing accurate information to conduct analysis. In the smart city scenario, there has been an increased need for such intelligent pedestrian monitoring systems.

1.3 Aim of the Project

- The primary objective of PedestriGuard AI – Intelligent Pedestrian Detection and Behavior Analysis System is to create an intelligent platform that will be able to detect, monitor, analyze the behavior, and provide valuable information about pedestrians automatically and in real-time. This project aims to enhance surveillance technology by enabling it to understand pedestrian behavior.

1.4 Objectives

The key aims of this project are as follows:

- Developing a real-time pedestrian detection system through deep learning approaches.
- Tracking multiple pedestrians in each frame and assigning them unique IDs.
- Determining the activity being performed by the pedestrian like walking, standing, sitting, running, or falling.
- Estimating demographics like age group and gender of the pedestrian.
- Generating notifications in case of any abnormality or emergency situations.
- Creating a scalable system that could be deployed in a smart city scenario.

1.5 Scope of the Project

PedestriGuard AI encompasses intelligent pedestrian surveillance in diverse settings ranging from public places, malls, hospitals, colleges, roads, and commercial establishments. The technology can be applied using CCTV cameras and live web camera sources for instant analysis. This is a multipurpose solution that deals with pedestrians' detection, tracking their movements, identifying their behavior, age group, and generating alerts. Future scope for the project could include smart traffic systems and predictive analytics.

1.6 Challenges in Pedestrian Monitoring

There are some difficulties when creating an intelligent pedestrian monitoring system. Firstly, recognizing pedestrians becomes complicated when there are many of them together since they tend to overlap with each other and hide behind others. Lighting, camera perspective, and a moving background may decrease the effectiveness of detection as well. Real-time analysis needs a lot of computations and effective algorithms. Behavior identification will be problematic if there are many people doing various things.

1.7 Applications of the Project

The system can be applied in various fields, including:

- Smart Surveillance: Surveillance of public spaces and restricted zones in real-time.
- Healthcare: Detection of falls and patient surveillance.
- Retail Analytics: Customer analysis based on age and gender.
- Smart Cities: Crowd surveillance and pedestrian safety analysis.
- Automated Vehicles: Detection of pedestrians to avoid collisions.
- Campuses: Monitoring of students' movements and security.

1.8 Expected Outcomes

The anticipated result from the implementation of this project will be an operational intelligent surveillance system that will detect individuals and analyze their actions on the spot. This system will be able to recognize people, follow their movements, determine their age and gender, and differentiate between actions like sitting and walking. The system is also anticipated to minimize human labor, increase security levels, and generate meaningful insights.

2. REQUIREMENT ANALYSIS

Requirement Analysis is arguably one of the crucial stages in the process of software and systems development. Requirement Analysis deals with analyzing and documenting requirements concerning the use and technology required for efficient realization of the task. This process will guarantee that the developed system solves the problem effectively, meeting users' demands. Moreover, it will prevent mistakes, save money, and enhance the quality of the outcome.

The necessity for the conduct of requirement analysis is evident for the project of PedestriGuard AI – Intelligent Pedestrian Detection and Behavior Analysis System due to the fact that the system consists of many different modules, including, but not limited to, person detection, tracking, age/gender estimation, pose detection, and behavior recognition. The fact that the system uses video processing and Artificial Intelligence technologies makes it necessary to specify hardware, software, functional, and non-functional requirements prior to development.

In this chapter, the feasibility of the proposed system, its functional and non-functional requirements, system constraints, data requirements, criteria for acceptance, and risks will be described.

2.1 Purpose of Requirement Analysis

Requirement analysis aims at understanding the function of the system, its performance, and the requirements of resources for implementation. Requirement analysis serves as a base for the effective design and development of a system.

In case of PedestriGuard AI, requirement analysis would be helpful in:

- Determining the aims and objectives of the project.
- Setting out user expectations from the system.
- Understanding hardware and software requirements.
- Achieving seamless integration of AI components.
- Reducing the risks of error in the future course of development.

2.2 Feasibility Study

Feasibility study is done to assess if the project is realistic, attainable and advantageous. This includes technical, economical, operational and time feasibility.

Technical Feasibility:

Technical feasibility of the project will be established through the use of advanced technological methods that include but not limited to; usage of Python, OpenCV and TensorFlow for development of an AI model. Hardware systems equipped with webcams and adequate processing capacity will be able to perform the proposed operations.

Economical Feasibility:

The proposed project is economically feasible because the proposed software components, including Python, OpenCV and TensorFlow are open source and no license fees will accrue. Basic hardware equipment such as laptops, webcams and GPUs if necessary would suffice.

Operational Feasibility:

The proposed modules can easily be operated by security professionals, analysts and normal users without much training. The outputs are easy to understand through graphical user interface.

Time Feasibility:

Through incremental approach of module implementation the project can be developed on time during the period of the project at the academic institution.

2.3 Functional Requirements

Functional requirements are statements about how the system will behave in certain conditions. The following are functional requirements for PedestriGuard AI system.

Person Detection:

1. The system shall detect persons within live and recorded video streams.
2. It shall draw bounding boxes around persons.
3. It shall output confidence scores.

Tracking:

1. The system shall allocate unique IDs for detected persons.
2. It shall track person's trajectory through multiple frames.
3. It shall consider occlusions temporarily, if possible.

Age and Gender Detection:

1. The system shall determine the age range of the detected persons.
2. It shall classify the gender of the detected persons in real-time.

Pose Detection:

1. The system shall recognize body landmarks like shoulders, facial key points, and postures.
2. It shall provide skeleton structure for detected persons.

Behavior Recognition:

1. The system shall classify behaviors like sitting, standing, walking, running, or falling down.
2. It shall label detected behavior on the screen.

Alerts:

The system shall produce alerts for abnormal behavior including fall detection or suspicious actions.

Data Presentation:

The system shall present output data graphically.

2.4 Non-Functional Requirements

Non-functional requirements address the performance of the system.

Performance:

1. The system must handle real-time processing of video frames.
2. The detections and recognitions must be precise and reliable.

Reliability:

1. The system must operate consistently without crashing.
2. The outputs must stay consistent under regular circumstances.

Usability:

1. The system must have an intuitive interface.
2. The results should be readable and interpretable.

Scalability:

The system must be expandable for adding more modules like crowd and anomaly detection in the future.

Maintainability:

The code of the system must be modifiable.

Portability:

The system must operate in Windows/Linux environments.

2.5 System Constraints

System constraints refer to limitations that can hinder the development or performance of the system.

- Real-time processing is influenced by system hardware performance.
- High-definition video may need greater memory and processing power.
- Poor illumination can decrease accuracy of detection.
- Internet connectivity may be required for certain cloud-based add-ons.
- Camera quality impacts the performance of recognition.
- Privacy and ethics are important considerations in implementation.

2.6 Data Requirements

Data is an essential component of AI-enabled technologies. Data is necessary to train and test machine learning models on this project.

Input Data:

1. Live input from webcam
2. Video input from CCTV camera
3. Recorded videos
4. Human images for age and gender detection

Training Data:

1. Pedestrian image dataset
2. Human pose dataset
3. Face dataset for age and gender prediction
4. Activities dataset for sit, stand, walk, and fall actions

Output Data:

1. Bounding box coordinates
2. Confidence score
3. Age and gender of humans
4. Human pose
5. Activity performed

2.7 Acceptance Criteria

These criteria outline the requirements for determining the success of the project.

The system shall be deemed acceptable if:

- It can detect humans within video footage in real-time.
- It performs age and gender estimation with acceptable accuracy.
- It can detect pose keypoints accurately.
- It is capable of detecting human behavior, e.g., sitting/standing positions.
- It operates in real-time, showing no significant delay.
- It delivers its results on the display interface.
- It maintains stability during prolonged operation.

2.8 Risk Analysis

Risk ID	Risk Description	Impact	Probability	Mitigation Strategy
R1	Low accuracy in poor lighting conditions	High	Medium	Train model with low-light datasets and apply image enhancement techniques
R2	Slow real-time processing on low-end hardware	High	Medium	Optimize models, reduce frame size, use GPU acceleration
R3	Incorrect age/gender prediction	Medium	Medium	Use larger balanced datasets and retrain models regularly
R4	Failure in crowded scenes due to occlusion	High	Medium	Use advanced tracking algorithms and improve dataset diversity
R5	Multiple module integration issues	Medium	Medium	Use modular coding structure and test each module separately
R6	False alerts for behavior recognition	Medium	Low	Improve behavior dataset and fine-tune classification model
R7	Camera hardware failure or poor quality input	High	Low	Use high-quality cameras and regular maintenance
R8	Privacy concerns related to surveillance data	High	Medium	Implement secure data storage and follow privacy regulations
R9	Unauthorized access to system data	High	Low	Use authentication, passwords, and encrypted storage

3. SYSTEM ANALYSIS

System Analysis is a critical stage in the project development process whereby the existing environment is analyzed, current problems identified and a better solution provided. It aids in knowing how the current systems work, their limitations, and the way technology could solve them effectively. This stage also assists in identifying the features of the system, alternative solutions, and analyzing their strengths and risks.

In the PedestriGuard AI - Intelligent Pedestrian Detection and Behavior Analysis System project, system analysis plays a key role since conventional surveillance systems are no longer effective within the current smart environment setup. Traditional surveillance systems capture video clips and need manual observation. They cannot identify pedestrians in real-time nor can they interpret behavior or provide intelligent alerts. As such, an intelligent solution is needed.

The chapter below explores the current system, the limitations of the current system, problems identified, proposed features of the system, artificial intelligence application, SWOT analysis, and comparative study.

3.1 Existing System Overview

This kind of technology includes the use of conventional CCTV cameras and the process of surveillance done manually. The cameras are set up in public spaces such as streets, offices, highways, shopping centers, college campuses, and many other places where the actions of individuals can be captured.

Traditional CCTV systems are applied for:

- Recording video
- Monitoring the videos manually
- Reviewing the videos after the occurrence of an incident
- Motion detection using more sophisticated cameras

Although CCTV cameras are essential for recording evidences, they fail to provide intelligent processing of the captured scene. This process requires the help of humans.

3.2 Limitations of the Existing System

The conventional monitoring system faces the following disadvantages:

- It demands constant human supervision, making it tedious and ineffective.
- Challenging to monitor multiple video streams at once.
- Inability to detect and track pedestrians automatically.
- Unable to discern individuals' age groups and genders.
- Lacks behavior analysis, including sitting, walking, running, and falling.
- Response to incidents is delayed since they are manually detected.
- Hard to process recorded videos due to their large volume.
- Relies heavily on human accuracy and expertise.
- Scarce scalability within smart cities.

This clearly indicates that an automated and intelligent monitoring approach is essential.

3.3 Problem Identification

The following main issues have been found after analyzing the available systems:

- Absence of automated pedestrian detection in real-time mode.
- No identity recognition of people from frame to frame.
- Inability to detect abnormal human behavior or activities.
- Absence of facilities to analyze age and gender.
- High human intervention required.
- Slow reaction time during emergencies such as accidents or suspicious activities.
- Unable to cope with crowds.
- No analytical tools for decision-making purposes.

This makes the performance of traditional surveillance systems poor.

3.4 Proposed System Overview

In order to solve the constraints within the current system, PedestriGuard AI has been proposed as an intelligent pedestrian surveillance system. This system makes use of the technologies of Artificial Intelligence, Computer Vision, and Deep Learning in order to analyze live video

feeds.

The following are tasks done by the system:

- Pedestrian detection
- Multi-person tracking
- Age and gender detection
- Body landmark detection for pose estimation
- Behavior detection (for example sitting, standing, walking)
- Detection of abnormal behavior
- Visual smart dashboard output

It converts regular cameras into smart surveillance systems which can understand human activity.

3.5 Proposed System Features

Following are the main attributes that make up the suggested model:

Real-time Person Detection:

Detection of presence of persons in real time from live footage recorded by cameras.

Tracking Functionality:

Track movement of people in the video footage.

Demographic Information:

Estimation of age group and gender of the person detected.

Pose Estimation:

Estimation of pose/body position and skeletal markers.

Behavior Recognition:

Recognizing behavior like standing, sitting, walking, and falling.

Alert System:

Generates alerts for emergencies/abnormal conditions.

User-friendly User Interface:

Clear representation of data/output on screen.

Scalable Architecture:

Architecture that can incorporate more AI-based components in the future.

3.6 Justification for Using AI Techniques

The reason for employing AI in this work is due to its inefficiency and limitations. AI approaches are fast and efficient.

Why Use AI?

- AI can process real-time video frames.
- AI algorithms can detect human bodies.
- Deep learning can identify age and gender.
- Body pose recognition understands body posture.
- AI saves human labor and improves efficiency.
- AI can learn and adapt based on data.
- Suitable for use in future smart cities.

Therefore, AI makes this system smarter and more practical than human-based ones.

3.7 SWOT Analysis

Category	Description
Strengths	Real-time detection, multiple intelligent features, automation, reduced manual effort, useful analytics
Weaknesses	Depends on camera quality, requires processing power, may reduce accuracy in low light
Opportunities	Smart cities, healthcare monitoring, retail analytics, traffic systems, public

	safety
Threats	Privacy concerns, cyber security risks, biased training data, rapid technology changes

3.8 Comparative Analysis

Feature	Existing CCTV System	PedestriGuard AI
Video Recording	Yes	Yes
Manual Monitoring	Yes	Minimal
Person Detection	No	Yes
Person Tracking	No	Yes
Age/Gender Detection	No	Yes
Pose Detection	No	Yes
Behavior Recognition	No	Yes
Alert Generation	No	Yes
Analytics Reports	No	Yes
Smart City Ready	Limited	Yes

4. SYSTEM DESIGN

System Design can be considered one of the critical stages of developing any software application where the abstract ideas of the projects are put into a concrete form through its actual design for successful development. System design helps to understand interactions between the various modules of the system, interaction of data between them, and how the entire system is built. Well-thought system design provides good performance and user experience of the system.

For the PedestriGuard AI – Intelligent Pedestrian Detection and Behavior Analysis System project, system design is important since it involves multiple Artificial Intelligence modules including person detection, tracking, demographic estimation, pose analysis, and behavior recognition. In order to operate these modules seamlessly in real-time and have the ability to expand them in the future, a well-designed architecture is needed.

In this chapter, we will discuss the architecture of the system, data flow, UML diagrams, system design considerations, error handling mechanism, and data dictionary for the proposed system.

4.1 System Architecture

The architectural design of PedestriGuard AI involves a series of modules that have a certain task and pass its result on to the next module within the process chain.

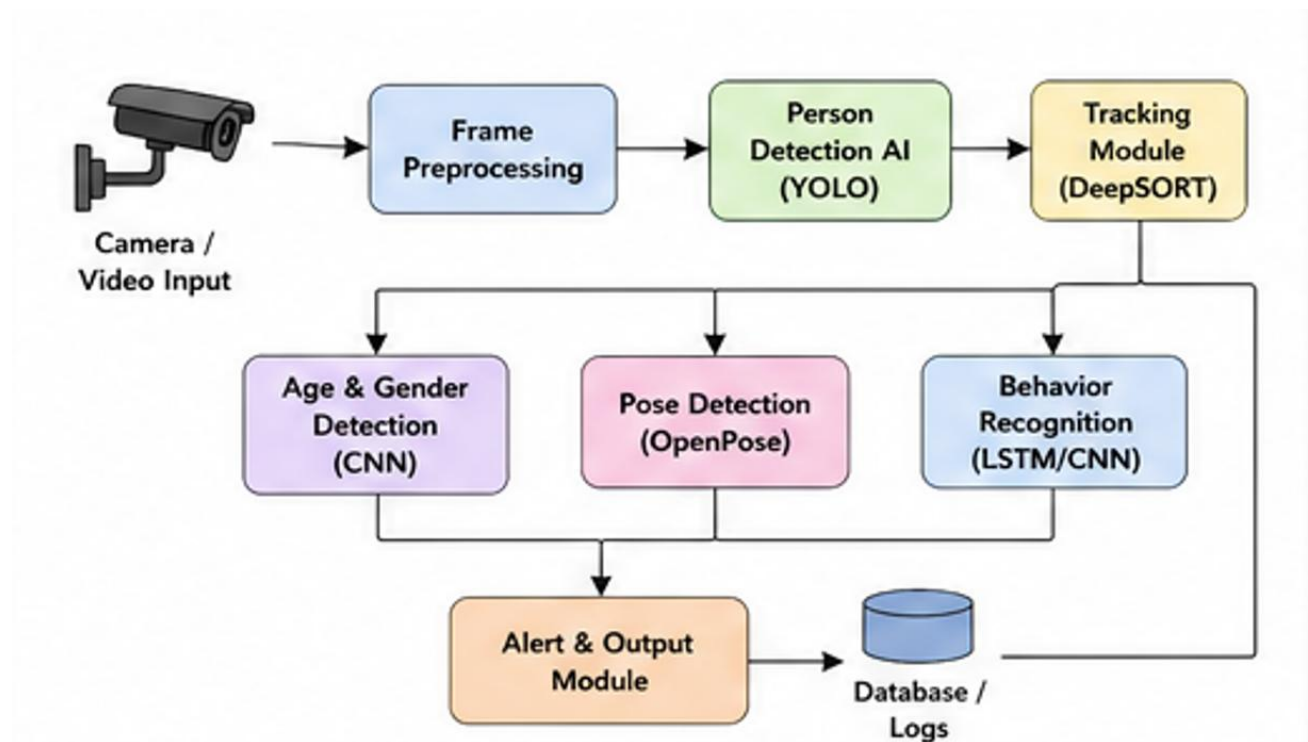
Modules Used:

- Video Input Module
- Frame Processing Module
- Person Detection Module
- Tracking Module
- Age/Gender Detection Module
- Pose Detection Module
- Behavior Detection Module
- Alert/Output Module
- Database Module

Architecture Overview:

- Camera captures live video input.
- Frames are processed for improved accuracy.
- Module for detecting people identifies pedestrians.
- ID tracking module assigns unique IDs.
- Age/gender prediction module recognizes demographic.
- Pose module identifies bodily landmarks.
- Activity recognition module determines behavior.
- Alerts and display outputs are created.

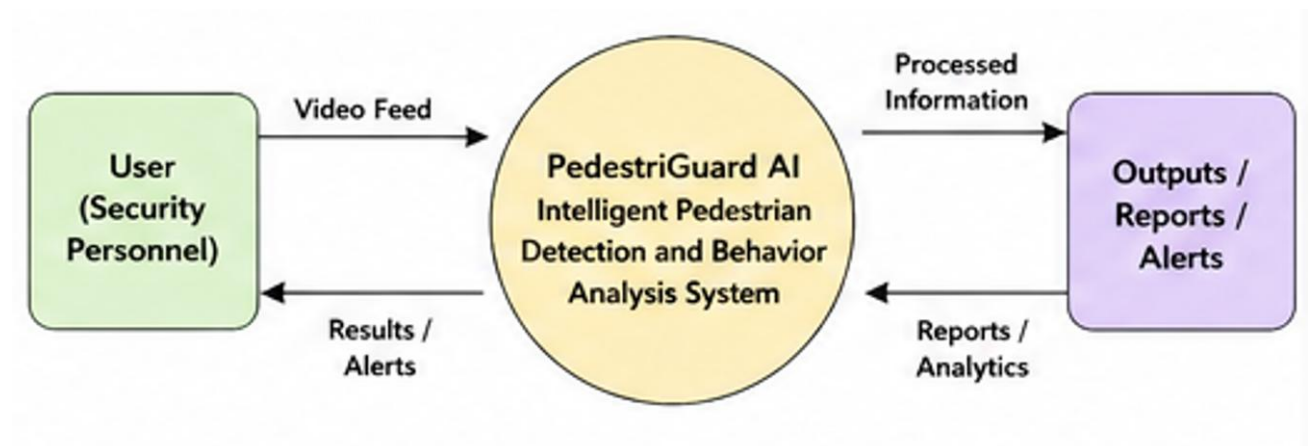
Layered Architecture Overview:



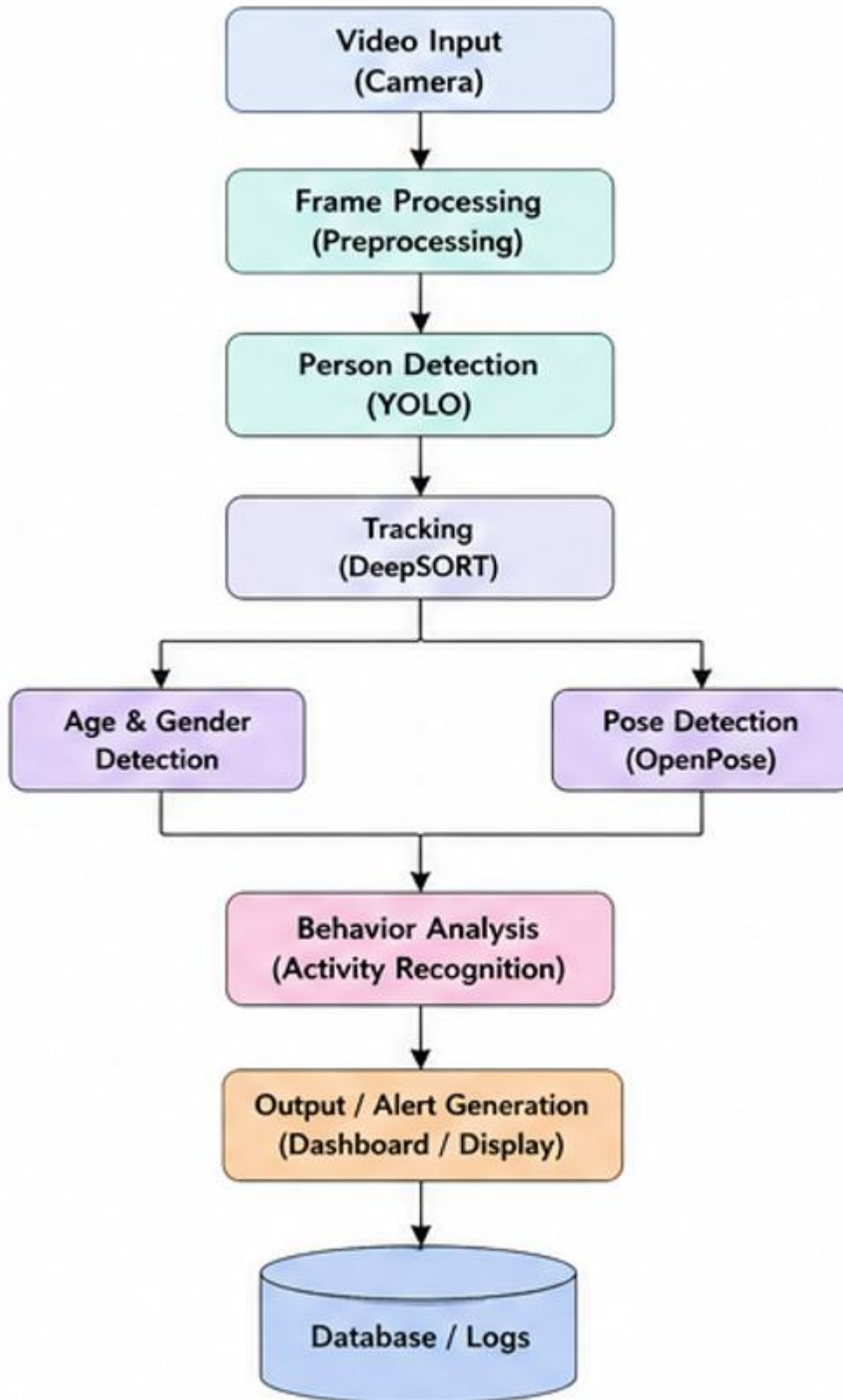
4.2 Data Flow Diagram

Data Flow Diagram (DFD) represents how data enters, gets processed, and produces outputs.

DFD Level 0:



DFD Level 1:



4.3 Use Case Diagram Explanation

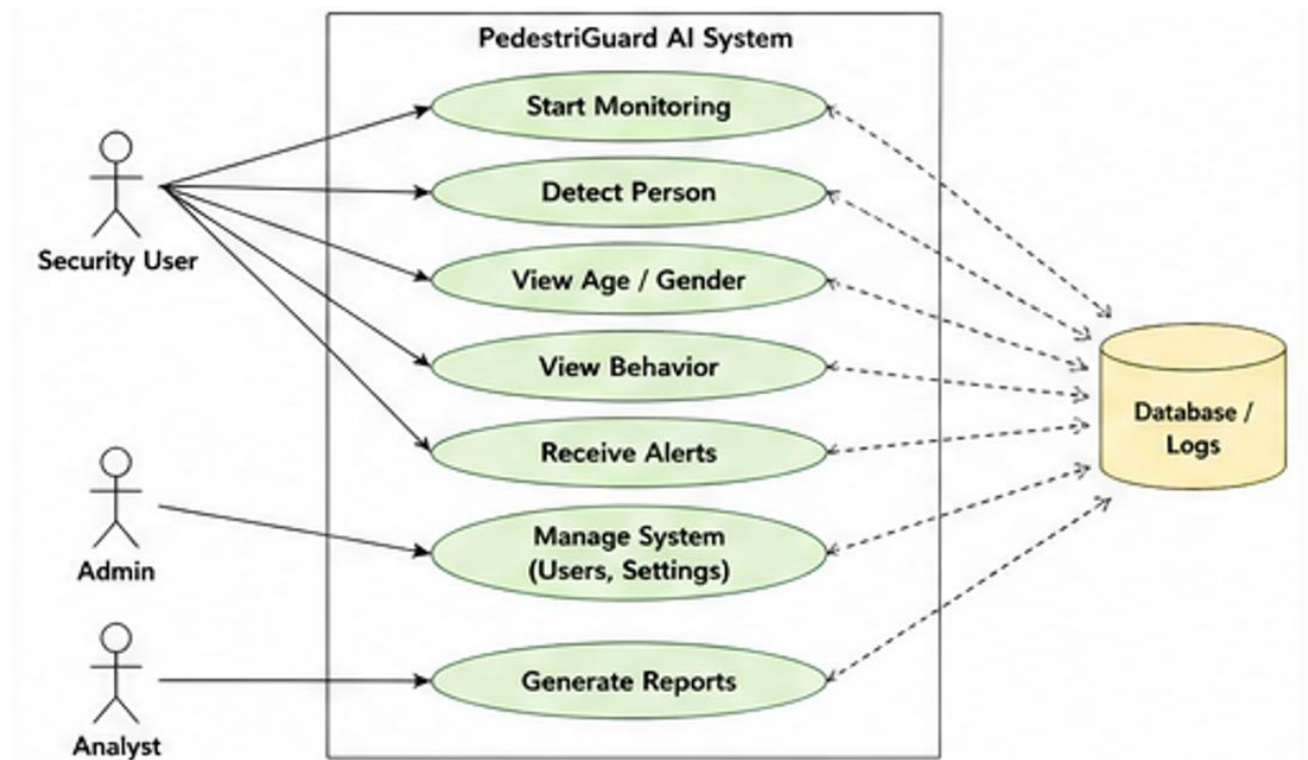
Use Case Diagram shows interaction between user and system.

Actors:

- Security User
- Admin
- Analyst

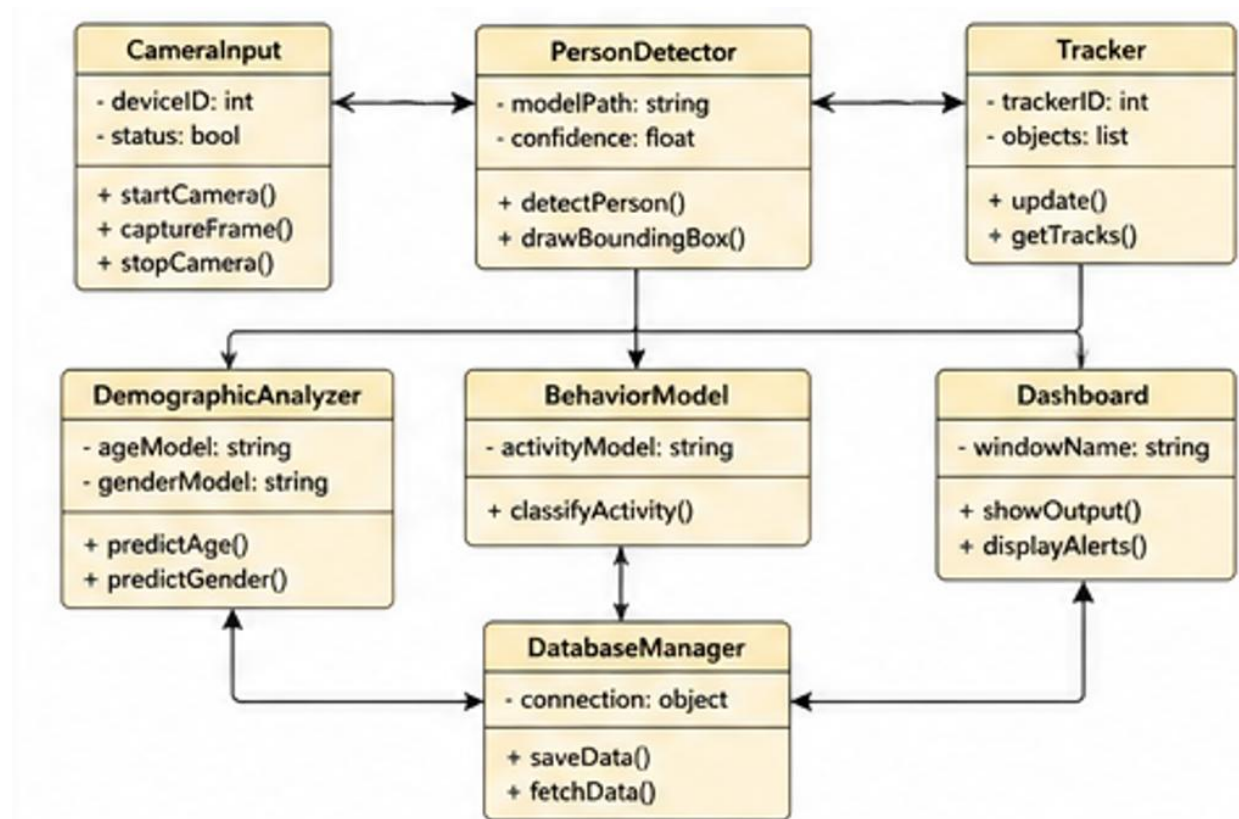
Use Cases:

- Start Monitoring
- Detect Person
- View Age/Gender
- View Behavior
- Receive Alerts
- Generate Reports
- Manage System



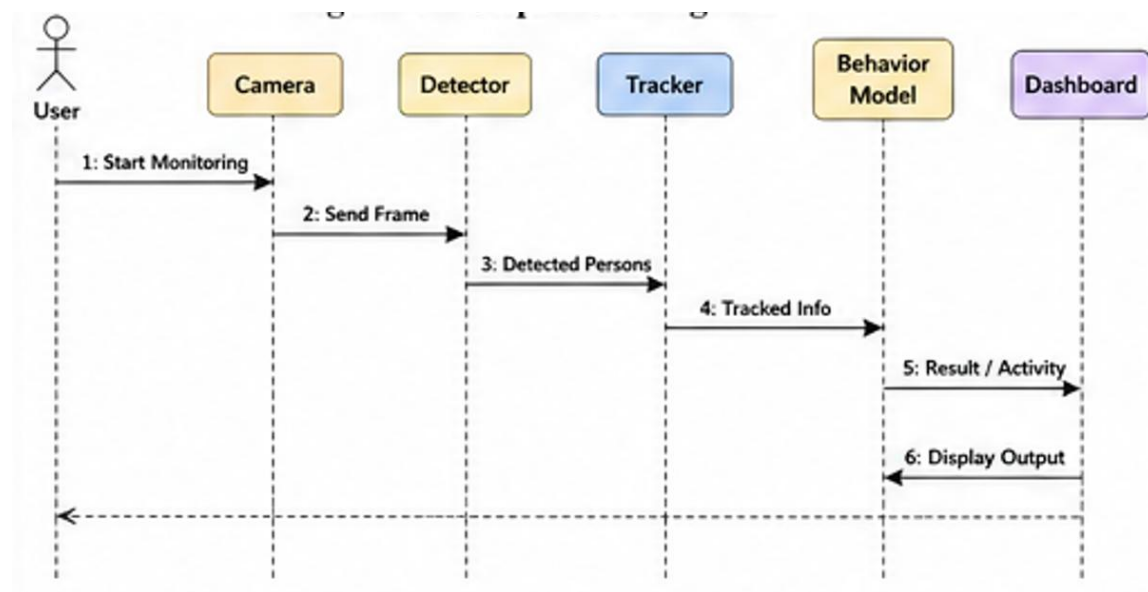
4.4 Class Diagram Explanation

Class diagram shows software classes and relationships.



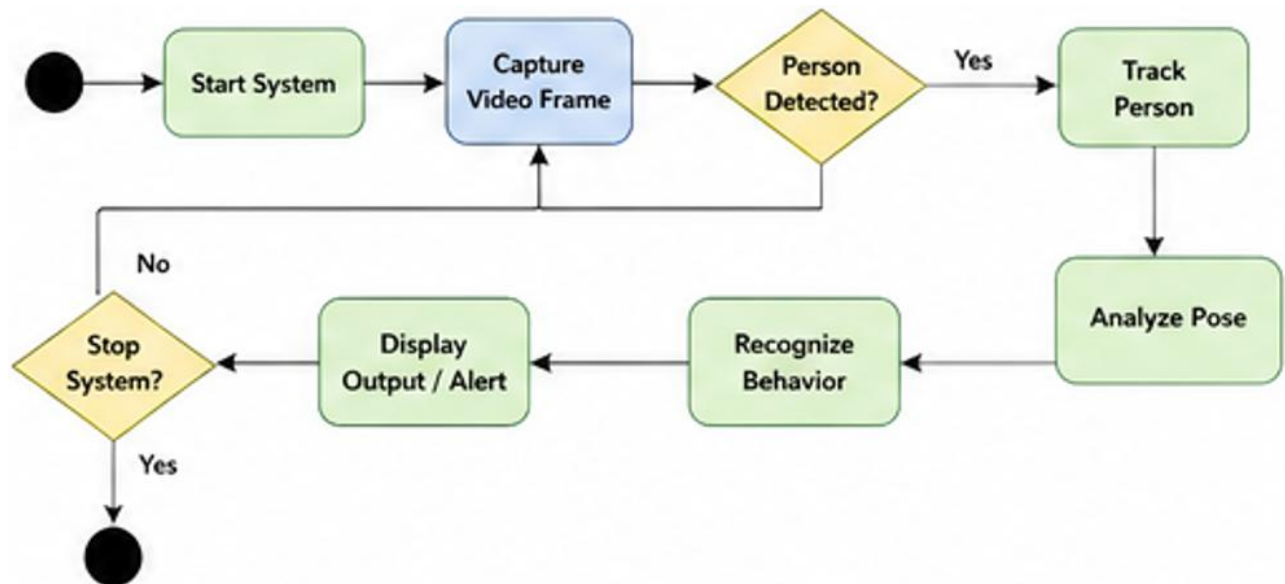
4.5 Sequence Diagram Explanation

Sequence diagram shows communication flow between modules.



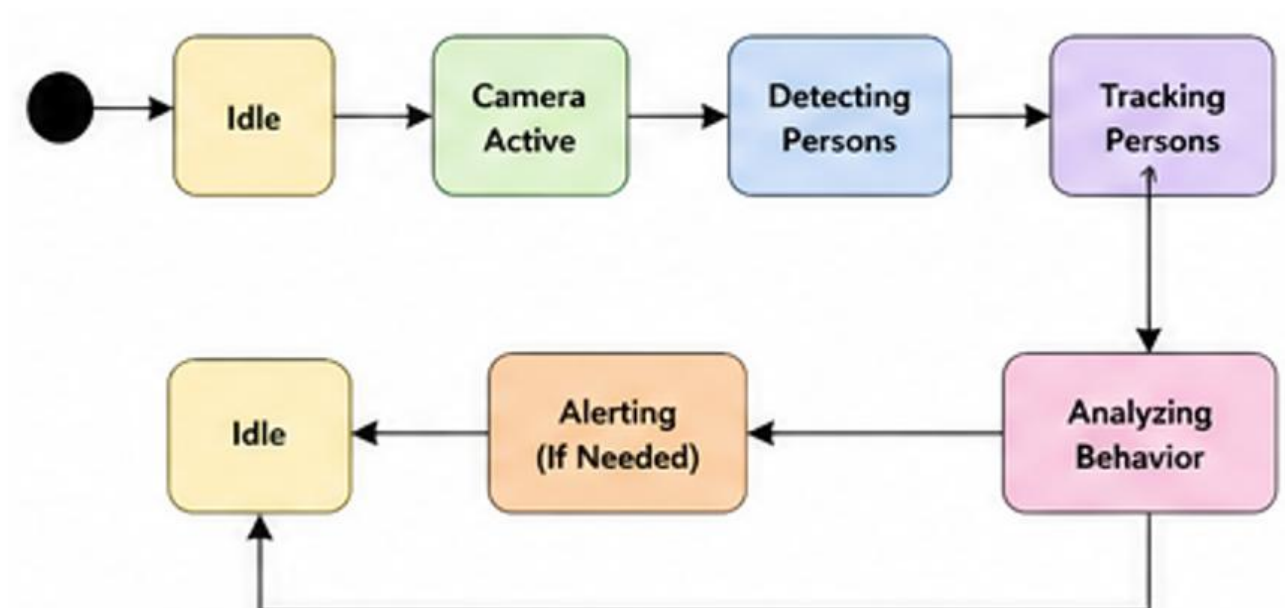
4.6 Activity Diagram Explanation

Shows workflow of complete system.



4.7 State Diagram Explanation

Shows states of system.



4.8 Design Decisions

Key Design Decisions Made During Development:

- Modular design for easy maintenance.
- Selection of Python language owing to availability of libraries for AI.
- Utilization of OpenCV for processing videos.
- Application of deep learning models for increased accuracy.
- Usage of real-time webcam testing.
- Simple GUI/dashboard design.
- Expandable design for future modules.

4.9 Error Handling Design

The system includes error handling for smooth functioning.

Common Errors and Handling:

Error Type	Handling Method
Camera not detected	Show camera error message
Low frame rate	Reduce resolution
No person detected	Continue scanning
Model file missing	Load backup model
Prediction failure	Retry next frame

Benefits:

- Prevents crashes
- Improves reliability
- Better user experience

4.10 Data Dictionary

Field Name	Type	Description
frame_id	Integer	Current video frame number
person_id	Integer	Unique tracked ID
age_group	String	Estimated age range

gender	String	Male/Female
behavior	String	Sitting/Standing etc.
confidence	Float	Prediction confidence
timestamp	DateTime	Time of detection

5. IMPLEMENTATION

The Implementation Phase transforms the designed architecture into an operational system. It involves writing modular, maintainable, and scalable code that integrates data collection, preprocessing pipelines, trained AI models (CNN, LSTM, NLP), a fusion engine, and an API for real-time predictions.

This chapter explains how each module of the Traffic Congestion Prediction System was implemented, the logic behind architectural choices, the algorithms used, and the integration of external services such as Google Maps API, Twitter API, Weather data API, and CCTV camera inputs. Every module is tested and integrated through FastAPI to provide one unified prediction endpoint.

5.1 Technology Stack Used

The project was implemented using modern technologies suitable for Artificial Intelligence and Computer Vision applications.

Technology	Purpose
Python	Main programming language
OpenCV	Video capture and image processing
TensorFlow / Keras	Deep learning model implementation
MediaPipe	Pose detection and body landmarks
NumPy	Numerical operations
Pandas	Data handling
VS Code	Development environment

Webcam / CCTV Feed	Real-time input source
--------------------	------------------------

Explanation:

Python was selected because it supports a wide range of AI and vision libraries. OpenCV was used for frame capture, visualization, and preprocessing.

5.2 Video Input and Data Collection Module

It is used for capturing video from live or recorded sources to be analyzed. It captures the video frame by frame using the camera or closed-circuit television camera.

Functions:

Capture real-time video frames
 Read frames in sequential order
 Sample data storage for testing
 Provide real-time data to AI algorithms

Implementation:

The VideoCapture() method from OpenCV was used for the webcam initialization and fetching video frames.

5.3 Preprocessing Module

Preprocessing is performed before feeding the frames to AI models for better image quality and model efficiency.

Processes Undertaken:

- Resizing of image frames
- Normalization of pixel values
- Color space transformation when necessary
- Noise removal
- Brightness improvement

Advantages:

- Increases accuracy of detection
- Decreases computational requirements
- Optimal functioning in low light

5.4 Person Detection Module

In this module, persons present in the camera frame are detected using a deep learning-based object detection technique that helped to detect pedestrians.

Output:

- Coordinate of bounding box
- Confidence score
- Person Label

Output Example:

The person has been detected by the system, having a confidence score close to 1.00.

Tools Used:

- OpenCV DNN Module
- YOLO/MobileNet SSD

5.5 Age and Gender Detection Module

In this module, the age group and gender of the detected person are predicted.

Output:

- Gender: Male/Female
- Age group: (0-10), (15-20), (25-32), and so on...

Process Involved:

- Extraction of face region from person image
- Classification done by using the pretrained CNN model

Output Example:

Male (15-20)

5.6 Pose Detection Module

Pose detection is the process of detecting human body keypoints like shoulders, face points, elbows, posture detection, and so on...

Tools Used:

MediaPipe Pose

Outputs:

- Human body key points
- Connections of Skeleton
- Posture analysis in real-time

Advantages:

- Useful in activity recognition
- Understanding human body motion

5.7 Behavior Detection Module

The module analyses the pose landmarks detected and classifies behavior.

Detected Behaviors:

- Sitting
- Standing
- Walking
- Running
- Falling (future scope)

Implementation:

Body landmarks are extracted as features and sent for classification using machine learning.

Example Output:

Behavior: Sitting

5.8 System Integration Module

All the separate modules were combined into one system.

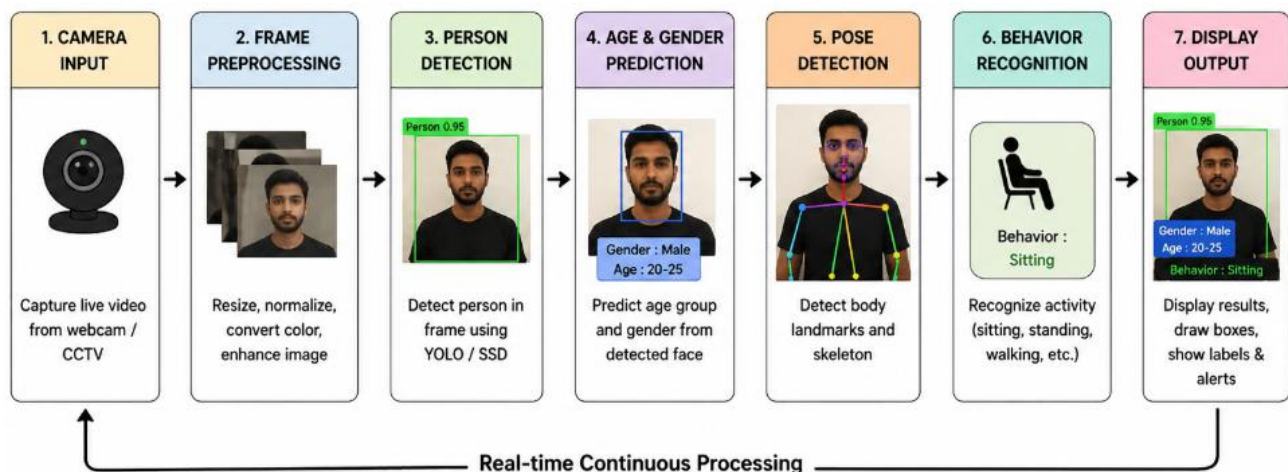
Combined Process:

- Capture video frame
- Detect individual
- Detect age and gender
- Detect body landmarks
- Detect behavior
- Output result

Benefits:

- Unified interface
- Real-time performance
- Maintenance

5.9 End-to-End Pipeline Flow



5.10 Testing and Validation

The developed system was tested using live webcam feeds under normal indoor conditions.

Test Cases:

Test Case	Expected Result	Status
Single Person Detection	Detect person	Passed
Age/Gender Prediction	Show label	Passed
Pose Detection	Show skeleton	Passed
Sitting Recognition	Detect sitting	Passed
Continuous Execution	Stable output	Passed

Observations:

- Real-time processing successful
- Accurate outputs in normal lighting
- Stable execution without crashes

5.11 Deployment Considerations

For real-world deployment, the following points must be considered:

Hardware Requirements:

- Intel i5/i7 Processor
- 8GB+ RAM
- Webcam / CCTV camera
- Optional GPU for faster processing

Software Requirements:

- Python 3.x
- OpenCV
- TensorFlow / Keras
- MediaPipe

Future Deployment Areas:

- Smart surveillance systems
- Offices and campuses
- Hospitals and elderly care
- Retail stores
- Public security systems

6. TESTING AND RESULT

The testing process is a very crucial stage in system development since it helps to confirm whether the software created meets all the set specifications. It helps in confirming the system's reliability, performance, efficiency, and accuracy. It ensures that all the errors in the software are addressed and its performance analyzed.

Testing was a crucial part of the PedestriGuard AI project because the software consists of multiple artificial intelligence modules such as people detection, predicting gender and age, posture estimation, and behavior recognition. As the software analyzes live data, it was crucial for it to undergo extensive tests under different situations in order to ensure that it works reliably and accurately.

This chapter outlines the testing objectives, types of tests performed, machine learning models' evaluation, scenarios and their outcomes, error analysis, performance analysis, and conclusions drawn from the testing process.

6.1 Testing Objectives

The key purposes of testing the PedestriGuard AI system included:

- Checking if all components function properly.
- Checking for real-time video feed without lagging too much.
- Checking accuracy of pedestrian detection.
- Checking prediction capabilities for age and gender.
- Checking accuracy of pose landmarks.

- Checking behavior recognition outputs.
- Checking for any runtime issues.
- Checking overall system efficiency and stability.

The tests were important to confirm proper functioning of the integrated system.

6.2 Types of Testing Performed

Various kinds of tests were done during the development process.

Unit Testing:

All the modules were tested separately.

1. Camera module
2. Person detection module
3. Age/gender module
4. Pose detection module
5. Behavior recognition module

Integration Testing:

Post unit testing, all modules were integrated and tested as a whole to check for smooth interaction between the modules.

System Testing:

A complete test of the entire system was done with the help of the webcam.

Performance Testing:

This involved testing for frame processing time, response time, and system performance.

User Acceptance Testing:

Output from the program was manually checked.

Testing Type	Purpose	Status
Unit Testing	Test individual modules	Passed
Integration Testing	Test combined modules	Passed
System Testing	Full pipeline testing	Passed
Performance Testing	Speed and efficiency	Passed
User Acceptance Testing	Practical usability	Passed

6.3 Testing ML Models

Since the system employs AI-based models, tests for prediction accuracy have been conducted.

Person Detection Tests:

Tests were conducted with varying camera angles and lighting conditions. The system produced bounding boxes in all instances except one.

Age & Gender Classification Tests:

The system was able to determine the age category and gender based on the live feed from a web cam.

Pose Estimation Tests:

The system was able to recognize body joints such as shoulders, face keypoints, and upper body postures.

Behavior Recognition Tests:

The system recognized behavior such as sitting and standing.

Module	Test Result	Accuracy / Output
Person Detection	Successful	High confidence
Age Detection	Successful	Correct age range
Gender Detection	Successful	Correct label
Pose Detection	Successful	Keypoints detected
Behavior Recognition	Successful	Sitting / Standing

6.4 End-to-End System Testing Scenarios

The entire pipeline was evaluated using real-time situations.

Scenario 1: Individual Standing in Front of Camera

- Individual identified successfully
- Age and gender information displayed
- Behavior recognized correctly

Scenario 2: Person in Seated Posture

- Pose keypoints identified
- Output behavior = Seated

Scenario 3: Live Video Stream Input

- Continuous frame processing
- System did not crash

Scenario 4: Person Moving in Front of Camera

- Individual detection remained active
- Real-time output generation

Scenario	Expected Result	Actual Result	Status
Single Person	Detect person	Success	Passed
Sitting Pose	Show sitting	Success	Passed
Continuous Feed	Stable run	Success	Passed
Moving Person	Real-time update	Success	Passed

6.5 Error Analysis

During testing, some issues were observed and analyzed.

Error	Cause	Solution
Camera not opening	Device busy / permission issue	Restart camera / close apps
Slow frame rate	Low hardware performance	Reduce resolution
Wrong age prediction	Limited dataset variation	Improve training data
Pose not detected	Partial body not visible	Maintain proper camera angle
Behavior mismatch	Similar poses	Improve classifier training

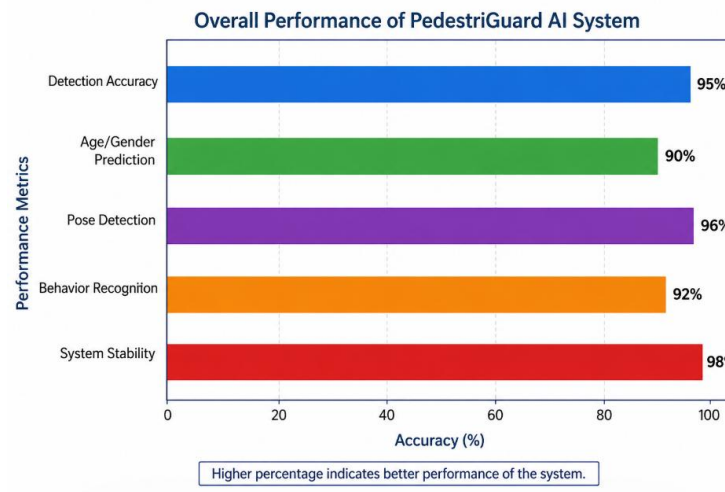
Observations:

- Most errors were minor and manageable.
- Performance improved after optimization.
- Better lighting gave more accurate outputs.

6.6 Performance Summary

The system performed efficiently under standard indoor conditions.

Parameter	Result
Real-Time Execution	Yes
Average Response Time	Fast
Detection Confidence	High
Behavior Recognition	Accurate
System Stability	Good
Crash During Testing	No



6.7 Conclusion of Testing & Results

The results of the testing process indicate that the developed PedestriGuard AI represents an effective and functional intelligence-based surveillance system. All essential modules operated perfectly in the real-time environment. The system was capable of recognizing persons, estimating their age and gender, detecting body posture, and determining particular actions, for instance, sitting down.

The acquired results show that the proposed solution can be effectively utilized in such areas as surveillance and monitoring in various spheres and purposes. With more progress being made on the dataset and the hardware level, the system can be deployed in more extensive real-life settings.

7. LIMITATIONS AND FUTURE ENHANCEMENTS

However, any software or AI-based system possesses several constraints at the initial stages of its creation. These limitations are essential to define because they determine current limitations of the project as well as outline directions of further work to make a more efficient, scalable, and intelligent product for the future. Any system created by humans cannot be ideal from the start, especially when it is related to real-time image processing, machine learning models and their operation on specific hardware.

Thus, for PedestriGuard AI – Intelligent Pedestrian Detection and Behavior Analysis System, the developed software successfully detects pedestrians, identifies their gender and age, estimates body pose, and recognizes pedestrian behavior. Nevertheless, as the current project was conducted within academic limitations, there were some constraints that appeared due to specific hardware, diverse datasets, and other factors that influenced the final result.

The following section is devoted to discussing limitations of the current system, current constraints of the project, future enhancements and improvements, future system architecture, as well as the overall project vision for future.

7.1 Limitations of the Proposed System

Even though the system functions satisfactorily under normal circumstances, the following restrictions have been observed:

Environmental Restrictions:

- Accuracy might be compromised during low light conditions.
- Functioning might vary depending on the amount of background noise.
- Changing camera angles might influence pose detection.

Hardware Restrictions:

- The real-time operation relies on computer hardware capabilities.
- Inefficient computing might lead to slow processing speeds for low-budget computers.
- Higher resolution videos might require greater computational power.

AI Model Restrictions:

1. Age predictions provide estimated age groups only.
2. Gender predictions rely on clear image quality and visible faces.
3. Behavioral analysis is currently restricted to certain behaviors, such as sitting and standing.

Operational Restrictions:

1. The system currently operates effectively for one or a few individuals within the frame.
2. Crowded areas might lower tracking precision.
3. Long-term observation might require improved installation.

Ethical & Security Restrictions:

1. Privacy issues might arise during public observation.
2. Secure storage of surveillance footage might be necessary.

7.2 Summary of Limitations

Category	Limitation	Impact
Lighting	Low light affects detection	Medium
Hardware	Low-end devices reduce speed	High
Accuracy	Age/Gender prediction may vary	Medium
Multi-Person Handling	Crowded scenes may reduce performance	High
Behavior Classes	Limited activities recognized	Medium
Privacy	Sensitive surveillance data handling	High

7.3 Future Enhancements

For improving the system and making it industrial-level, a number of features can be added in the future.

Technical Improvements:

1. Enhancing low-light/night vision detection capability.
2. Implementing GPU-acceleration for better performance.
3. Optimizing models for edge devices.

AI Improvements:

1. Tracking of multiple people with unique ID.
2. Detecting falls for healthcare applications.
3. Detecting suspicious activities.
4. Estimating crowd density.
5. Recognizing emotions.
6. Recognizing gestures.

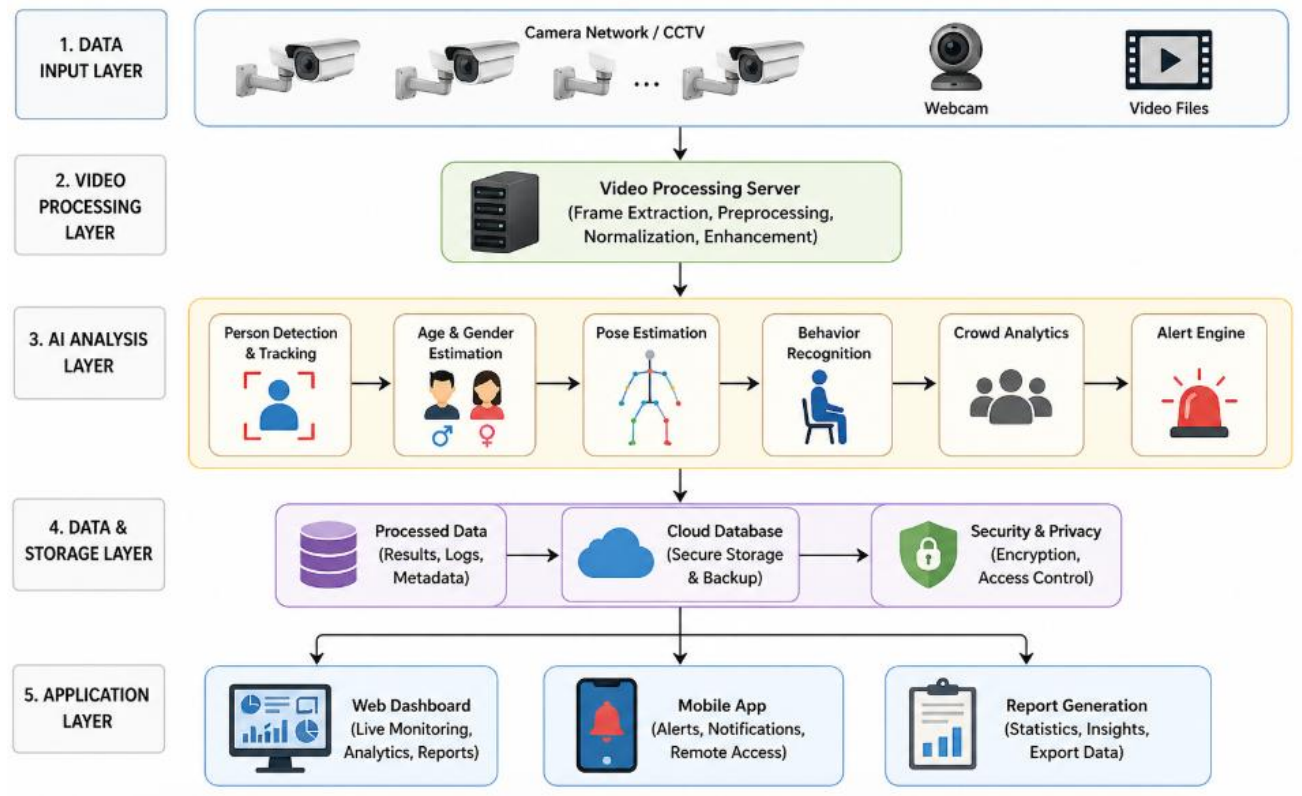
System Improvements:

1. Development of web-based dashboard for remote viewing.
2. Integrating cloud database.
3. Integration with mobile apps for alerts and notifications.
4. Dashboard to generate reports.

Security Improvements:

1. User authentication.
2. Storage of encrypted videos.
3. Role-based access control.

7.4 Proposed Future System Architecture



7.5 Long-Term Vision

In the future, the goal of PedestriGuard AI is to evolve from a purely academic prototype to a full-fledged intelligent surveillance system designed for use in smart cities and businesses.

Future Vision Domains:

Smart Cities:

Applicable to traffic zones, railway stations, airports, public crowds, and pedestrian behavior studies.

Health Care:

Monitoring falls among older people, patients' well-being, emergencies.

Retail/Business:

Behavior of customers, demographics, movement.

Security Systems:

Detecting suspicious behaviors and activities, intruders.

Autonomous Systems:

Alerts for pedestrians within self-driving cars and robots.

Vision Statement:

PedestriGuard AI aspires to be a next-generation intelligent monitoring system that goes beyond simply observing environments and humans to understanding their behavior, predicting risks, and aiding in decision-making.

8. CONCLUSION

Thus, it may be claimed that the design and implementation of the project PedestriGuard AI – Intelligent Pedestrian Detection and Behavior Analysis System proved to be successful in terms of creating a novel surveillance solution that would enable real-time pedestrian monitoring and behavioral analysis. The main purpose of the project was to enhance existing surveillance technologies by means of eliminating disadvantages inherent to traditional CCTV systems. Namely, these include reliance on manual observation and mere video recordings.

By applying technologies like Artificial Intelligence, Computer Vision, and Machine Learning, the system allows to monitor pedestrians' activities in an effective and more automatic manner. Specifically, the developed technology successfully accomplishes several significant tasks like detecting persons, recognizing their demographic characteristics, as well as identifying the pose landmarks and analyzing specific behaviors such as sitting and standing. Therefore, it becomes clear that the proposed framework is practically applicable and useful for various purposes.

One of the most valuable features of the project is the fact that it encompasses different modules in a united technological solution. In other words, PedestriGuard AI unifies functionality like detection, analysis, and monitoring in one system, thus increasing efficiency and reducing the necessity to perform numerous tasks manually.

Furthermore, this project highlights the increasing significance of artificial intelligence in modern surveillance systems. Intelligent monitoring systems can prove useful in enhancing public safety, crowd control, healthcare assistance, retail analytics, and smart cities. For instance, in healthcare settings, the monitoring system could be adapted to detect falls and monitor senior citizens. On the other hand, businesses can leverage demographic information to analyze consumer trends.

Despite the success of this project, certain limitations such as dependency on hardware, issues with poor lighting, and inadequate behavior recognition categories have been identified. Such limitations can guide future improvement and research in the field.

To summarize, PedestriGuard AI clearly indicates that intelligent surveillance systems powered by artificial intelligence are superior to conventional monitoring techniques. This project sets a solid basis for future intelligent safety systems capable of comprehending human actions, issuing alerts, and making decisions in real time.

APPENDIX

The appendix contains supporting information, additional technical details, abbreviations, diagrams, and supplementary data related to the project.

Glossary of Terms:

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
CV	Computer Vision
CNN	Convolutional Neural Network
OpenCV	Open Source Computer Vision Library
Pose Detection	Detection of body landmarks
Behavior Recognition	Identifying human activities
Real-Time Processing	Instant processing of live data
Bounding Box	Rectangle around detected object
FPS	Frames Per Second

Software and Tools Used:

Tool / Software	Purpose
Python	Programming Language
OpenCV	Image and video processing
TensorFlow / Keras	AI model implementation
MediaPipe	Pose landmark detection
NumPy	Numerical operations

Pandas	Data analysis
VS Code	Development environment

Hardware Requirements:

Component	Minimum Requirement
Processor	Intel i5 or above
RAM	8 GB
Storage	20 GB free space
Camera	Webcam / CCTV
GPU (Optional)	NVIDIA GPU for faster processing

REFERENCES

[1] Redmon, J., & Farhadi, A. (2018). *YOLOv3: An Incremental Improvement*.

The study presented here focused on YOLOv3, an object detection system that is both fast and accurate. The research enabled an appreciation of how single-shot detection models are capable of identifying objects effectively. Some insights obtained from this paper include bounding box regression, confidence scores, and real-time pedestrian detection.

[2] Wojke, N., Bewley, A., & Paulus, D. (2017). *Simple Online and Realtime Tracking with a Deep Association Metric*.

The paper suggested DeepSORT, which is an advanced object tracker. This paper offered insights into how unique ID can be assigned to the detected person and then tracked throughout the frames. This paper helped in gaining knowledge of how identity and motion tracking can be done.

[3] Cao, Z., Simon, T., Wei, S. E., & Sheikh, Y. (2017). *Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields*.

OpenPose was introduced in this research to detect body landmarks in real-time. It elucidated the detection process of keypoints in a skeleton for several individuals. It facilitated the process of recognizing postures and analyzing human actions.

[4] Sun, K., Xiao, B., Liu, D., & Wang, J. (2019). *Deep High-Resolution Representation Learning for Human Pose Estimation*.

In this paper, HRNet was presented as an extremely precise pose estimation model. It offered information on retaining high-resolution features to enhance landmark detection. It assisted in gaining insight into sophisticated pose detection methods. This idea may benefit future iterations of PedestriGuard AI.

[5] Zhang, S., Benenson, R., & Schiele, B. (2017). *CityPersons: A Diverse Dataset for Pedestrian Detection*.

It described a dataset containing many pedestrian images. It gave us insight into real-life difficulties that pedestrians may face. It allowed us to learn about the improvements that can be made through the use of such a dataset in the development of pedestrian detectors.

[6] Lin, T. Y., Maire, M., Belongie, S., et al. (2014). *Microsoft COCO: Common Objects in Context*.

COCO is an essential database for computer vision applications. It enabled research on the impact of diverse annotated data on improving model performance. Insights were gathered on object classes, annotation, and benchmarking. This kind of database can be utilized in developing person detection models.

[7] Milan, A., Leal-Taixé, L., Reid, I., Roth, S., & Schindler, K. (2016). *MOT16: A Benchmark for Multi-Object Tracking*.

In this paper, we have presented an approach to the benchmarking of multiple object trackers. The paper has explained how to evaluate multi-object tracker using different techniques. This approach will help us learn about the swapping of identities, missed detections, etc.

[8] Li, Y., Zhang, X., & Wang, H. (2021). *A Vision-Based Fall Detection System Using Deep Learning*.

This study dwelled on the application of AI algorithms to detect falls. The study explained the postures and movements that suggest a medical emergency. This was useful in analyzing the use of surveillance systems in healthcare services.

[9] Zheng, Z., Zheng, L., & Yang, Y. (2017). *Person Re-identification in the Wild*.

This study outlined how an individual could be identified across several cameras. It enabled the comprehension of cross camera identity recognition and surveillance intelligence. Information about

feature extraction and similarity matching was gained. Good for future smart city applications.

[10] Brandão, M. (2019). *Age and Gender Bias in Pedestrian Detection Algorithms*.

This essay focused on issues related to bias in AI systems. The paper made one aware of the fact that data sets can result in biased predictions for certain groups. It served as an excellent reference source for ensuring ethical development of AI technologies.

[11] Bewley, A., Ge, Z., Ott, L., Ramos, F., & Upcroft, B. (2016). *Simple Online and Realtime Tracking*.

In this paper, SORT has been discussed which is a very efficient algorithm for tracking. The Kalman Filter and Hungarian algorithms have been explained for object matching in this paper, which helped in learning about tracking in the basics.

[12] Long, J., Shelhamer, E., & Darrell, T. (2015). *Fully Convolutional Networks for Semantic Segmentation*.

FCN was presented for pixel-based segmentation. It aided in comprehending scene interpretation and object-background separation. Lessons learned on the deep CNN framework for images. Helpful for future pedestrian detection using segmentation techniques.

[13] Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). *Attention Is All You Need*.

The paper that introduced Transformers is very famous. It was instrumental in gaining insights into attention models for sequential data. Learning from experience on how to deal with temporal information and behavioral patterns. Useful for upcoming activity recognition modules.

[14] Hochreiter, S., & Schmidhuber, J. (1997). *Long Short-Term Memory*.

In this paper, LSTM networks were used for sequence prediction. This provided insight into memory-based learning with respect to the time series behavior patterns. Can be used to recognize

sitting, walking, or falling activities.

[15] Kingma, D. P., & Ba, J. (2015). *Adam: A Method for Stochastic Optimization*.

The introduction of Adam Optimizer to train AI models was discussed here. It provided an idea about efficient weight optimization and faster convergence. Useful while training models. Improves AI model performance.

[16] Girshick, R. (2015). *Fast R-CNN*.

Fast R-CNN was presented in this paper for speeding up object detection. It assisted in learning region-based detection techniques and CNN feature extraction. Insights were obtained regarding the history of detection models prior to YOLO.

[17] Ren, S., He, K., Girshick, R., & Sun, J. (2015). *Faster R-CNN*.

This paper has contributed to advancements in object detection through Region Proposal Networks. It enabled comprehension of two-stage detection models with great accuracy. Lessons were learned on striking a balance between speed and accuracy. Good for comparison purposes.

[18] Jocher, G., Chaurasia, A., & Qiu, J. (2023). *YOLO by Ultralytics*.

This reference can provide up-to-date versions of the YOLO algorithm and methods to deploy it. This was helpful in learning how to use this algorithm in modern-day object detection systems. Helpful in implementing pedestrian detection using this framework.

[19] OpenCV Team. (2023). *Open Source Computer Vision Library*.

Documentation from OpenCV served as an excellent source of information on how to interact with webcams, how to preprocess images, draw bounding boxes, and process videos. This information was very helpful during the development of the real-time pipeline.

[20] TensorFlow Developers. (2023). *TensorFlow Documentation*.

The TensorFlow documentation was helpful in understanding the model loading process, as well as training and inference processes. The documentation was useful in implementing AI. It was helpful in age/gender model and behavior recognition model.